

BIOINFORMATICS BOOTCAMP 2026

Assessment 01: Biological Databases & Foundations

QUESTION 1

A researcher has discovered a DNA sequence that may be associated with a rare genetic disorder. Before conducting laboratory experiments, they want to determine whether this sequence has been reported previously. Which biological database(s) would you recommend for this purpose? Justify your answer by explaining the type of information each database provides and how it would assist the researcher.

To verify if a newly discovered human DNA sequence has been previously reported or linked to a disease, I would recommend using NCBI GenBank paired with BLASTn (Nucleotide BLAST), alongside ClinVar and OMIM (Online Mendelian Inheritance in Man).

- NCBI GenBank / BLASTn: The researcher can input the novel sequence into BLASTn to search GenBank's comprehensive primary repository. This determines if an identical or highly similar nucleotide sequence already exists in the global literature.
- ClinVar & OMIM: If the sequence matches an existing gene, searching ClinVar will reveal known assertions regarding its pathogenicity and clinical significance (whether it is classified as benign, pathogenic, or a variant of uncertain significance). Concurrently, OMIM provides a detailed, curated overview of human genes and their specific relationships to genetic disorders, helping the researcher establish a concrete link between the sequence and the rare disease.

QUESTION 2

A scientist is studying a disease-causing bacterium and wants to compare its genome with those of related bacterial species to identify conserved genes. Which biological database(s) would you recommend for this analysis? Explain why these databases are appropriate and what information they provide.

For comparative prokaryotic genomics aimed at finding conserved regions, the best resources are the NCBI RefSeq (Reference Sequence) database and Ensembl Bacteria.

- NCBI RefSeq: Unlike primary archives, RefSeq provides non-redundant, highly curated, and well-annotated reference genomes. It allows the scientist to download standard, high-quality reference genomes of related bacterial strains to serve as a baseline for comparison.
- Ensembl Bacteria: This platform is specifically designed for visualizing and analyzing bacterial clades. It offers interactive comparative genomics tools, such as pre-calculated whole-genome alignments and orthologue predictions.

By running a comparative tool like BLASTp or using Ensembl's orthology features against these databases, the scientist can efficiently isolate genes that remain unchanged across multiple pathogenic species, highlighting critical survival mechanisms or potential drug targets.

QUESTION 3

A biotechnology company has isolated a protein from a newly identified microorganism and wants to investigate whether similar proteins exist in other species. Which biological database(s) would you recommend, and what information would you expect to obtain from each database? Explain your reasoning.

To search for homologous proteins across different taxa, I recommend using the UniProtKB (Universal Protein Knowledgebase)—specifically UniProtKB/TrEMBL and UniProtKB/Swiss-Prot—combined with a BLASTp (Protein BLAST) search against the NCBI non-redundant (nr) protein database.

- UniProtKB/Swiss-Prot & TrEMBL: This is the most comprehensive resource for protein sequences. Swiss-Prot provides manually annotated, high-quality entries with validated functional data, while TrEMBL covers automatically annotated sequences, ensuring that even uncharacterized proteins from recently sequenced organisms are included.
- NCBI non-redundant (nr) database: Running a BLASTp search against this repository scans all known translated coding regions across all registered species.

Inputting the query protein sequence into BLASTp against these databases will yield a list of matching sequences ranked by E-value and percentage identity. This directly shows the company whether close orthologs exist in other species, helping them map out the evolutionary distribution and functional conservation of the protein.

QUESTION 4

A scientist has identified an unknown protein sequence and wants to determine its function. Which biological database(s) would you recommend for this purpose? Justify your answer by explaining the type of information each database provides.

When a protein sequence is unknown, standard sequence alignment might fail if the evolutionary distance is too great. Therefore, I recommend a multi-layered approach using UniProtKB/Swiss-Prot, InterPro (incorporating Pfam), and the PDB (Protein Data Bank).

- UniProtKB/Swiss-Prot: A direct BLASTp search here might instantly reveal a characterized match with pre-documented biochemical functions, saving significant analytical time.
- InterPro / Pfam: If there are no direct whole-sequence matches, InterPro classifies the sequence into protein families and predicts functional domains or important active sites by scanning signatures from databases like Pfam. Identifying a conserved domain (like a kinase domain or a DNA-binding motif) tells the scientist exactly what the protein does biochemically.

- Protein Data Bank (PDB) & AlphaFold DB: Searching structural repositories can reveal 3D structural homologs. Because protein structure is more evolutionarily conserved than sequence, finding a structural match provides strong clues about its functional mechanisms.

QUESTION 5

You have been provided with an unknown biological sequence but are not told whether it is DNA, RNA, or protein. Describe the step-by-step approach you would follow to identify the sequence using biological databases. Explain which databases you would use at each stage and why.

To identify a completely mysterious biological sequence, a systematic computational pipeline must be executed:

- Step 1: Alphabet and Composition Analysis: First, check the characters making up the sequence string. If the sequence contains characters beyond A, C, T, G, or U (such as M, F, W, Y, L), it is immediately identified as a protein sequence. If it only contains A, C, G, T, it is likely DNA. If it contains U instead of T, it is RNA.
- Step 2: Database Search for Nucleotide Queries: If identified as a nucleotide sequence (DNA/RNA), run a BLASTn search against the NCBI nucleotide (nt) database. This will match the sequence against all known genomic, transcriptomic, and synthetic sequences to identify the exact organism of origin and the specific gene or locus.
- Step 3: Translation and Cross-Validation: If no direct nucleotide match is found, or if the sequence is highly divergent, run a BLASTx search (which translates a nucleotide query in all 6 reading frames against a protein database) or a standard BLASTp (if identified as a protein in Step 1) against UniProtKB. Protein sequences are much more conserved than nucleotide sequences; thus, a protein-level search will successfully identify functional homologs, revealing what protein it codes for and its biological classification.